NIOS Class 10 (212) Science & Technology — Top 50 Most Repeated Objective Questions

NIOS Secondary (Class 10) Science & Technology (212) — Exam Prep Study Kit

NIOS Class 10 (212) Science & Technology — Top 50 Most Repeated Objective Questions

Master the 42 Objective Marks in Section A (MCQs, True/False & Blanks)

> **Section A carries 42 marks out of 85.**
> Scoring 35+ on these questions guarantees a passing grade before you even write a single descriptive answer in Section B!

Part 1: Multiple Choice Questions (1 Mark Each)

1. Physics (Optics, Mechanics & Electricity)

Q1. The SI unit of electric potential difference is:
(A) Ampere
(B) Volt
(C) Ohm
(D) Watt

> **Answer: (B) Volt**
> *Explanation*: $V = W/Q$ (Joules per Coulomb = Volt).

Q2. The commercial unit of electrical energy is kilowatt-hour (kWh). $1\ \mathrm{kWh}$ is equal to:
(A) $3.6 \times 10^5\ \mathrm{J}$
(B) $3.6 \times 10^6\ \mathrm{J}$

(C) 36×10^6 J
(D) 3.6×10^4 J

> **Answer: (B)** 3.6×10^6 J
> *Explanation*: $1\ \text{kW} \times 1\ \text{hr} = 1000\ \text{W} \times 3600\ \text{s} = 3.6 \times 10^6\ \text{J}$.

Q3. A person cannot see distant objects clearly, but can see nearby objects distinctly. The defect of vision he is suffering from is:
(A) Hypermetropia
(B) Myopia
(C) Presbyopia
(D) Cataract

> **Answer: (B) Myopia**
> *Explanation*: Myopia is short-sightedness, corrected with a concave lens.

Q4. A lens has a power of $+2.0\ \text{D}$. Its focal length is:
(A) $+50\ \text{cm}$
(B) $-50\ \text{cm}$
(C) $+20\ \text{cm}$
(D) $+100\ \text{cm}$

> **Answer: (A) +50 cm**
> *Explanation*: $f = \frac{1}{P} = \frac{1}{+2} = +0.5\ \text{m} = +50\ \text{cm}$ (Convex lens).

Q5. If the resistance of a wire of length l is R, then another wire of the same material having double the length and same cross-sectional area will have resistance:
(A) $R/2$
(B) $2R$
(C) $4R$
(D) R

> **Answer: (B) 2R**
> *Explanation*: $R \propto l$. Doubling length doubles the resistance.

Q6. The acceleration due to gravity on the surface of the Moon is approximately:
(A) $\frac{1}{2}$ that of Earth
(B) $\frac{1}{6}$ that of Earth
(C) 6 times that of Earth
(D) Equal to that of Earth

> **Answer: (B) 1/6 that of Earth**

Q7. An object of mass $10\ \text{kg}$ is taken to the Moon. Its mass on the Moon will be:
(A) $10\ \text{kg}$
(B) $\frac{10}{6}\ \text{kg}$
(C) $60\ \text{kg}$
(D) $0\ \text{kg}$

> **Answer: (A) 10 kg**
> *Explanation*: Mass is constant everywhere. (Weight changes to $\frac{1}{6}$, but mass remains $10\ \text{kg}$).

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2. Chemistry (Acids, Salts, Metals & Carbon)

Q8. The chemical formula of Bleaching Powder is:
(A) CaCl_2
(B) CaOCl_2
(C) Ca(OH)_2
(D) CaSO_4

> **Answer: (B)** CaOCl_2 (Calcium oxychloride)

Q9. Which of the following salts is used for supporting fractured bones in medicine?
(A) Bleaching powder
(B) Baking soda
(C) Plaster of Paris
(D) Washing soda

> **Answer: (C) Plaster of Paris** $(\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O})$

Q10. The pH of an aqueous solution is measured to be 3. The solution is:
(A) Strongly alkaline
(B) Weakly alkaline
(C) Strongly acidic
(D) Neutral

> **Answer: (C) Strongly acidic** (pH < 7 is acidic; lower pH means higher acidity).

Q11. A metal which is liquid at room temperature is:
(A) Sodium
(B) Mercury
(C) Bromine
(D) Gallium

> **Answer: (B) Mercury (Hg)**
> *(Note: Bromine is a liquid NON-metal; Mercury is a liquid METAL)*.

Q12. Which of the following metals is stored under kerosene oil to prevent accidental fire?
(A) Magnesium
(B) Aluminium
(C) Sodium
(D) Iron

> **Answer: (C) Sodium (Na)** (and Potassium).

Q13. The non-metal which is a good conductor of electricity is:
(A) Diamond
(B) Graphite
(C) Sulphur
(D) Phosphorus

> **Answer: (B) Graphite** (Allotrope of carbon with free delocalized electrons).

Q14. The functional group present in an alcohol molecule is:
(A) $-\mathrm{CHO}$
(B) $-\mathrm{COOH}$

(C) $-\mathrm{OH}$
(D) $>\mathrm{C}=\mathrm{O}$

> **Answer: (C) -OH**

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3. Biology & Environment

Q15. The basic structural and functional filtration unit of the human kidney is:
(A) Neuron
(B) Nephron
(C) Alveolus
(D) Glomerulus

> **Answer: (B) Nephron**

Q16. In humans, fertilization of the egg by a sperm normally takes place in the:
(A) Uterus
(B) Ovary
(C) Fallopian tube (Oviduct)
(D) Vagina

> **Answer: (C) Fallopian tube (Oviduct)**

Q17. According to Lindeman's 10% law, if $1000\ \mathrm{J}$ of solar energy is trapped by producers, the energy available to secondary consumers will be:
(A) $100\ \mathrm{J}$
(B) $10\ \mathrm{J}$
(C) $1\ \mathrm{J}$
(D) $0.1\ \mathrm{J}$

> **Answer: (B) 10 J**
> *Explanation*: Producers $(1000\ \mathrm{J}) \rightarrow$ Herbivores $(100\ \mathrm{J}) \rightarrow$ Carnivores $(10\ \mathrm{J})$.

Part 2: Fill in the Blanks (2 Marks Each / 1 Mark Per Part)

Q18.

(i) Newton's first law of motion is also known as the law of _____.

> **Answer: inertia**

(ii) The product of mass and velocity of a body is called _____.

> **Answer: momentum** ($p = mv$)

Q19.

(i) The device used to measure electric current in a circuit is called _____.

> **Answer: ammeter**

(ii) An ammeter is always connected in _____ in an electrical circuit.

> **Answer: series** (Voltmeter is connected in parallel)

Q20.

(i) The focal length of a plane mirror is _____.

> **Answer: infinity**

(ii) The power of a concave lens is always _____.

> **Answer: negative**

Q21.

(i) The chemical formula of Baking Soda is _____.

> **Answer:** $\mathrm{NaHCO_3}$ (Sodium hydrogen carbonate)

(ii) Heating Gypsum at $373\ \mathrm{K}$ produces _____.

Answer: Plaster of Paris ($\mathrm{CaSO_4 \cdot \frac{1}{2}H_2O}$)

Q22.

(i) The property of self-linking of carbon atoms through covalent bonds is known as _____.

Answer: catenation

(ii) The general formula of Alkanes is _____.

Answer: $\mathrm{C_nH_{2n+2}}$

Q23.

(i) The hormone secreted by the pancreas that regulates blood sugar level is _____.

Answer: insulin

(ii) The plant hormone responsible for phototropism (bending towards light) is _____.

Answer: auxin

Q24.

(i) In Mendel's monohybrid cross, the phenotypic ratio in the F_2 generation is _____.

Answer: $3:1$

(ii) Normal human somatic cells contain _____ pairs of chromosomes.

Answer: 23 (46 chromosomes total)

Q25.

(i) The gas responsible for depleting the stratospheric ozone layer is _____.

Answer: Chlorofluorocarbon (CFC)

(ii) The progressive accumulation of non-biodegradable toxic substances along a food chain is termed _____.

> **Answer: biomagnification**

Part 3: True / False Statements (2 Marks Each / 1 Mark Per Part)

Q26.

(i) "Weight of an object remains constant everywhere in the universe."

> **Answer: FALSE**
> *Reason*: Mass remains constant; weight depends on the local acceleration due to gravity ($W = mg$).

(ii) "Action and reaction forces act on two different bodies."

> **Answer: TRUE**
> *Reason*: Newton's Third Law states forces act on different bodies in opposite directions.

Q27.

(i) "The resistivity of a metallic wire depends on its length and thickness."

> **Answer: FALSE**
> *Reason*: Resistance (R) depends on length and area, but resistivity (ρ) is an intrinsic material property depending only on the nature of the material and temperature.

(ii) "An electric fuse wire should have high resistance and a low melting point."

> **Answer: TRUE**
> *Reason*: So that it melts quickly and breaks the circuit in case of an overload.

Q28.

(i) "A convex mirror always forms a virtual, erect, and diminished image."

> **Answer: TRUE**
> *Reason*: This is why convex mirrors are used as rear-view mirrors in vehicles.

(ii) "Red light bends the most when passing through a glass prism."

> **Answer: FALSE**
> *Reason*: Violet light bends the most; red light has the longest wavelength and bends the least.

Q29.
(i) "Pure water has a pH equal to 7 at $25^{\circ}\mathrm{C}$."

> **Answer: TRUE**

(ii) "Acids turn red litmus paper blue."

> **Answer: FALSE**
> *Reason*: Bases turn red litmus blue; acids turn blue litmus red.

Q30.
(i) "Saponification is the chemical reaction used to prepare soaps."

> **Answer: TRUE**

(ii) "Diamond and graphite are chemical isomers of carbon."

> **Answer: FALSE**
> *Reason*: They are allotropes, not isomers.

Q31.
(i) "Deoxygenated blood enters the left atrium of the human heart."

> **Answer: FALSE**
> *Reason*: Deoxygenated blood from the body enters the RIGHT atrium; oxygenated blood enters the LEFT atrium.

(ii) "In humans, the male gamete determines the sex of the unborn child."

> **Answer: TRUE**
> *Reason*: The father donates either an X or Y chromosome, whereas the mother always donates an X.

Q32.

(i) "Energy flow in an ecosystem is unidirectional."

> **Answer: TRUE**
> *Reason*: Energy flows from the Sun $\rightarrow$ producers $\rightarrow$ herbivores $\rightarrow$ carnivores and cannot flow backward.

(ii) "Biodegradable substances are the primary cause of biomagnification."

> **Answer: FALSE**
> *Reason*: Non-biodegradable chemicals like DDT cause biomagnification because they cannot be broken down.